COMPUTER NETWORK: PHYSICAL LAYER

ASSIGNMENT 1

Sapna Baranwal

STACKPULSE

March 13, 2026

Outline

1 MULTIPLE CHOICE QUESTIONS (MCQs)

MCQ Q21: Last Host of Subnet – Question

Question MCQ Q21

Question: Given the address 210.69.92.39/26, determine the last usable host address in the second-to-last subnet when subnets are arranged in ascending order.

(A) 210.69.92.127/26

(B) 210.69.92.192/26

(C) 210.69.92.191/26

(D) 210.69.92.254/26

MCQ Q21: Last Host of Subnet – Strategy

Problem-Solving Strategy

- /26 means block size 64 addresses.
- 210.69.92.39 is in subnet 210.69.92.0/26.
- Subnets: .0, .64, .128, .192.
- Second-to-last subnet = .128/26.
- Last host of .128 subnet = .190, broadcast = .191.

MCQ Q21: Last Host of Subnet – Solution

Solution

Answer: (C) 210.69.92.191/26

The broadcast address of the second-to-last subnet is 210.69.92.191.

MCQ Q22: Subnet ID and Host ID – Question

Question MCQ Q22

Question: For the IP address 141.14.196.46 with subnet mask 255.255.192.0, what are the subnet identifier and host identifier?

(A) 141.14.192.0 and 141.14.4.46

(B) 141.14.1.46 and 141.14.192.0

(C) 255.255.192.0 and 255.255.4.46

(D) None of the above

MCQ Q22: Subnet ID and Host ID – Strategy

Problem-Solving Strategy

- IP: 141.14.196.46 = 141.14.11000100.00101110
- Mask: 255.255.192.0 = 255.255.11000000.00000000
- Subnet ID (AND): 141.14.11000000.00000000 = 141.14.192.0
- Host ID (IP & complement): complement = 0.0.00111111.11111111
- IP & complement = 0.0.00000100.00101110 = 141.14.4.46

MCQ Q22: Subnet ID and Host ID – Solution

Solution

Answer: (A) 141.14.192.0 and 141.14.4.46

Subnet ID = 141.14.192.0, Host ID = 141.14.4.46.

MCQ Q23: CIDR Notation – Question

Question MCQ Q23

Question: An organization receives 2048 consecutive addresses from 128.211.160.0 to 128.211.191.255. Express this block in CIDR notation.

(A) 128.211.168.0/11

(B) 128.211.160.0/29

(C) 128.211.160.0/19

MCQ Q23: CIDR Notation – Strategy

Problem-Solving Strategy

- Block size = 2048 addresses = $2^{11} \rightarrow 11$ host bits.
- Network bits = 32 - 11 = 21 (theoretical).
- Verify by finding common bits between first and last address.
- First: 128.211.160.0 = 10000000.11010011.10100000.00000000
- Last: 128.211.191.255 = 10000000.11010011.10111111.11111111
- Common bits = first two octets (16) + first 3 bits of third octet = 19.

MCQ Q23: CIDR Notation – Solution

Solution

Answer: (C) 128.211.160.0/19

First 19 bits are common, so CIDR = /19.

MCQ Q24: IP Address Classes – Question

Question MCQ Q24

Question: Identify the correct address classes for the following IPs:
X = 202.23.14.150, Y = 168.19.200.12, Z = 72.192.52.210

(A) X is Class A, Y is Class B, Z is Class C

(B) X is Class C, Y is Class A, Z is Class B

(C) X is Class C, Y is Class B, Z is Class A

(D) X is Class A, Y is Class C, Z is Class B

MCQ Q24: IP Address Classes – Strategy

Problem-Solving Strategy

- Class A: 1-126
- Class B: 128-191
- Class C: 192-223
- $X = 202 \rightarrow$ Class C
- $Y = 168 \rightarrow$ Class B
- $Z = 72 \rightarrow$ Class A

MCQ Q24: IP Address Classes – Solution

Solution

Answer: (C) X is Class C, Y is Class B, Z is Class A

Based on first octet ranges.

MCQ Q25: Class A Network Address – Question

Question MCQ Q25

Question: A host has IP 25.34.12.56 with subnet mask 255.255.0.0. What is the network address of its subnet?

(A) 25.34.12.0

(B) 25.34.12.56

(C) 25.34.0.0

(D) 25.0.0.0

MCQ Q25: Class A Network Address – Strategy

Problem-Solving Strategy

- Network address = IP & Mask (bitwise AND).
- IP: 25.34.12.56 = 00011001.00100010.00001100.00111000
- Mask: 255.255.0.0 = 11111111.11111111.00000000.00000000
- AND result: first two octets remain, last two become 0.
- Result: 00011001.00100010.00000000.00000000 = 25.34.0.0

MCQ Q25: Class A Network Address – Solution

Solution

Answer: (C) 25.34.0.0

Bitwise AND gives 25.34.0.0.

MCQ Q26: Class B Network Address – Question

Question MCQ Q26

Question: A host has IP 125.134.112.66 with subnet mask 255.255.224.0. What is the network address?

(A) 125.134.96.0

(B) 125.134.112.0

(C) 125.134.112.66

(D) 125.134.0.0

MCQ Q26: Class B Network Address – Strategy

Problem-Solving Strategy

- IP: 125.134.112.66 = 01111101.10000110.01110000.01000010
- Mask: 255.255.224.0 = 11111111.11111111.11100000.00000000
- Third octet AND: 01110000 & 11100000 = 01100000 = 96
- Result: 01111101.10000110.01100000.00000000 = 125.134.96.0

MCQ Q26: Class B Network Address – Solution

Solution

Answer: (A) 125.134.96.0

Bitwise AND gives 125.134.96.0.

MCQ Q27: First Host of First Subnet – Question

Question MCQ Q27

Question: In a Class C network, a computer has IP 200.99.39.112 and subnet mask 255.255.255.224. What is the last octet value of the first usable host in the very first subnet?

MCQ Q27: First Host of First Subnet – Strategy

Problem-Solving Strategy

- Mask 255.255.255.224 = 11100000 → 3 subnet bits, 5 host bits.
- IP last octet 112 = 01110000 → subnet bits = 011 (subnet 3).
- First subnet (subnet 1) has subnet bits = 001.
- First host of subnet 001: host bits all 0 except last = 1 → 00100001.
- 00100001 binary = 33 decimal.

MCQ Q27: First Host of First Subnet – Solution

Solution

Answer: 33

First host of first subnet = 200.99.39.33, so last octet = 33.

MCQ Q28: Last Host Identification – Question

Question MCQ Q28

Question: With subnet mask 255.255.255.224, which of these could be the last host address in some subnet?
(i) 210.15.16.62, (ii) 210.15.16.94, (iii) 210.15.16.127, (iv) 210.15.16.191

(A) (i) and (ii) only

(B) (i) and (iii) only

(C) (ii) and (iv) only

(D) (iii) and (iv) only

MCQ Q28: Last Host Identification – Strategy

Problem-Solving Strategy

- Mask 255.255.255.224 $\rightarrow$ block size = 32.
- Last host pattern: host bits = 11110 (30 in decimal).
- (i) 62 = last octet 00111110 $\rightarrow$ host bits 11110 $\rightarrow$ last host
- (ii) 94 = last octet 01011110 $\rightarrow$ host bits 11110 $\rightarrow$ last host
- (iii) 127 = last octet 01111111 $\rightarrow$ host bits 11111 $\rightarrow$ broadcast
- (iv) 191 = last octet 10111111 $\rightarrow$ host bits 11111 $\rightarrow$ broadcast

MCQ Q28: Last Host Identification – Solution

Solution

Answer: (A) (i) and (ii) only

62 and 94 are last hosts; 127 and 191 are broadcast addresses.

MCQ Q29: Addressable Hosts – Question

Question MCQ Q29

Question: A network is identified by the subnet ID 134.96.51.64 with a 26-bit prefix. How many devices can be directly addressed in this network (excluding the network and broadcast addresses)?

MCQ Q29: Addressable Hosts – Strategy

Problem-Solving Strategy

- /26 means 26 network bits, 6 host bits.
- Total addresses $= 2^6 = 64$.
- Reserved: network ID and broadcast address.
- Usable hosts $= 64 - 2 = 62$.

MCQ Q29: Addressable Hosts – Solution

Solution

Answer: 62

$2^6 - 2 = 62$ addressable hosts.

MCQ Q30: Maximum Data Rate – Question

Question MCQ Q30

Question: In a network with maximum packet size of 600 bytes, packet lifetime of 30 seconds, and 8-bit sequence numbers, calculate the maximum possible data rate in bits per second for a single connection.

MCQ Q30: Maximum Data Rate – Strategy

Problem-Solving Strategy

- 8-bit sequence number $\rightarrow$ max distinct packets $= 2^8 = 256$.
- In 30 seconds, at most 256 packets can be sent.
- Total data in 30s $= 256 \times 600$ bytes.
- Convert to bits: $256 \times 600 \times 8$.
- Divide by time: $(256 \times 600 \times 8)/30$.

MCQ Q30: Maximum Data Rate – Solution

Solution

Answer: 40960 bps

$(256 \times 600 \times 8)/30 = 1,228,800/30 = 40,960$ bps.

MCQ Q31: Feasible Subnet Set – Question

Question MCQ Q31

Question: A company has Class C network 200.200.200.0 and needs four subnets: two with 60 hosts each, one with 20 hosts, and one with 10 hosts. Which option shows a valid allocation of subnet addresses and masks?

(A) 200.200.200.64/26, 200.200.200.128/26, 200.200.200.32/27, 200.200.200.16/28

(B) 200.200.200.128/25, 200.200.200.128/25, 200.200.200.192/27, 200.200.200.240/28

(C) 200.200.200.192/25, 200.200.200.128/26, 200.200.200.96/28, 200.200.200.102/28

(D) 200.200.200.128/25, 200.200.200.192/25, 200.200.200.64/26, 200.200.200.208/27

MCQ Q31: Feasible Subnet Set – Strategy

Problem-Solving Strategy

- 60 hosts $\rightarrow$ need /26 (255.255.255.192), block size 64.
- 20 hosts $\rightarrow$ need /27 (255.255.255.224), block size 32.
- 10 hosts $\rightarrow$ need /28 (255.255.255.240), block size 16.
- Subnets must be contiguous and non-overlapping.
- Option (A): .64/26, .128/26, .32/27, .16/28 $\rightarrow$ non-overlapping, fits in /24.

MCQ Q31: Feasible Subnet Set – Solution

Solution

Answer: (A)

Option (A) provides non-overlapping subnets with correct mask sizes.

MCQ Q32: $X^2 + Y^2$ Calculation – Question

Question MCQ Q32

Question: For the Class C address 200.25.80.117, let X = number of 1-bits in its network ID, and Y = number of 1-bits in its directed broadcast address. Compute $X^2 + Y^2$.

MCQ Q32: $X^2 + Y^2$ Calculation – Strategy

Problem-Solving Strategy

- Class C $\rightarrow$ network ID = 200.25.80.0, DBA = 200.25.80.255.
- Count 1's in each octet:
- 200 = 11001000 $\rightarrow$ 3 ones
- 25 = 00011001 $\rightarrow$ 3 ones
- 80 = 01010000 $\rightarrow$ 2 ones
- 0 = 00000000 $\rightarrow$ 0 ones
- 255 = 11111111 $\rightarrow$ 8 ones
- X = 3+3+2+0 = 8
- Y = 3+3+2+8 = 16
- $X^2 + Y^2 = 64 + 256 = 320$

MCQ Q32: $X^2 + Y^2$ Calculation – Solution

Solution

Answer: 320

$X = 8$, $Y = 16$, $X^2 + Y^2 = 64 + 256 = 320$.

MCQ Q33: Max IP Not Assignable – Question

Question MCQ Q33

Question: In the network 251.12.10.110/28, what is the fourth octet value of the highest address that cannot be assigned to any host?

MCQ Q33: Max IP Not Assignable – Strategy

Problem-Solving Strategy

- /28 means 28 network bits, 4 host bits $\rightarrow$ block size $= 16$.
- 110 in binary $= 01101110$.
- First 4 bits (network bits) $= 0110 = 6$.
- Subnet address $= 251.12.10.96/28$ (since $01100000 = 96$).
- Address range: .96 to .111.
- .96 = network ID, .111 = broadcast address.
- Broadcast cannot be assigned to host $\rightarrow$ fourth octet $= 111$.

MCQ Q33: Max IP Not Assignable – Solution

Solution

Answer: 111

Subnet is 251.12.10.96/28, broadcast is 251.12.10.111, so fourth octet = 111.